


# 32D Series — Metal Oxide Varistors

32 mm round, epoxy-encapsulated MOVs with radial leads · Datasheet · Rev. 2026-10-21

## Description

The 32D series is a line of 32 mm round, epoxy-encapsulated zinc oxide varistors made by Guiyang High-Tech Yige Electronic Co., Ltd. for surge protective device (SPD) manufacturers and OEMs. Rated In 20 kA / Imax 40 kA (8/20 µs), the standard range covers varistor voltages from 100 V to 1,200 V in 27 model grades. The varistor discs are pressed, sintered, metallized, coated and tested in our own Guiyang plant — not bought in — which is how tolerances and surge ratings stay consistent from batch to batch. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.



## Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- 32 mm round MOVs, epoxy encapsulated, with radial leads
- 27 model grades — varistor voltage from 100 V to 1,200 V
- In 20 kA, Imax 40 kA (8/20 µs)
- Low-residual-voltage range available for tighter clamping
- Leaded disc format for PCB and point-to-point assembly
- Discs pressed, sintered, metallized and tested in our own plant

## Agency approvals

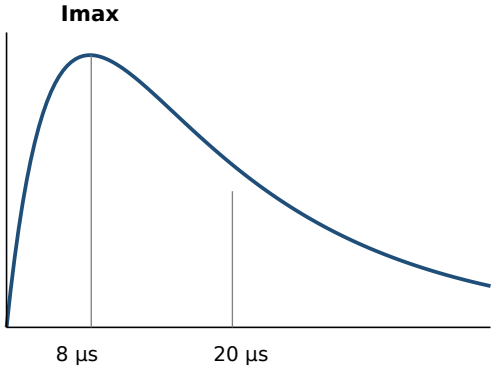
Column definitions: U1mA = varistor voltage at DC 1 mA; ΔU1mA = varistor-voltage tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In, shown as reference value–ratio; ETM = maximum energy absorption (2 ms). The catalog prints UL / C-UL / TÜV columns for this table but leaves them blank for every model in this series, so they are omitted here.

For series not yet UL recognized, we can pursue UL certification for your program. Full certification coverage for all series is on the Certifications page at www.flyboatmov.com.

## Absolute maximum ratings

| Parameter | Range / limit | Unit |
|---|---|---|
| Continuous operating voltage, AC (VM(AC)rms) | 60 to 750 | V |
| Continuous operating voltage, DC (VM(DC)) | 85 to 1060 | V |
| Nominal discharge current In (8/20 µs) | 20 | kA |
| Maximum discharge current Imax (8/20 µs) | up to 40 | kA |
| Maximum energy absorption (2 ms) | up to 1400 | J |

**Impulse current waveform 8/20 µs (definition)**




CAUTION: stresses above the ratings in the tables below may cause permanent damage. The ranges above summarize the per-model ratings in the sections below; always design to the row for the exact model code.

## Model naming and standard marking

Model naming: 32D K 431 — 32 = nominal size (mm), D = round disc; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; -U = uncoated bare electrode disc (no mark = epoxy-coated by default). Standard marking: the manufacturer logo and the model code are etched on the encapsulation.

## 1. Electrical characteristics — standard ratings

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

| Model | U1mA (V) | ΔU1mA (%) | UAC (V) | UDC (V) | In (kA) | Imax (kA) | UP (V) | ETM (J) |
|---|---|---|---|---|---|---|---|---|
| 32DK101 | 100 | ±10 | 60 | 85 | 20 | 40 | 360-3.6 | 150 |
| 32DK121 | 120 | ±10 | 75 | 100 | 20 | 40 | 400-3.6 | 185 |
| 32DK151 | 150 | ±10 | 95 | 125 | 20 | 40 | 450-3.0 | 230 |
| 32DK181 | 180 | ±10 | 115 | 150 | 20 | 40 | 490-2.8 | 270 |
| 32DK201 | 200 | ±10 | 130 | 170 | 20 | 40 | 520-2.6 | 310 |
| 32DK221 | 220 | ±10 | 140 | 180 | 20 | 40 | 550-2.6 | 350 |
| 32DK241 | 240 | ±10 | 150 | 200 | 20 | 40 | 600-2.6 | 370 |
| 32DK271 | 270 | ±10 | 175 | 225 | 20 | 40 | 680-2.6 | 430 |
| 32DK301 | 300 | ±10 | 195 | 250 | 20 | 40 | 770-2.6 | 480 |
| 32DK331 | 330 | ±10 | 210 | 270 | 20 | 40 | 830-2.55 | 520 |
| 32DK361 | 360 | ±10 | 230 | 300 | 20 | 40 | 910-2.55 | 580 |
| 32DK391 | 390 | ±10 | 250 | 320 | 20 | 40 | 1050-2.55 | 640 |
| 32DK431 | 430 | ±10 | 275 | 350 | 20 | 40 | 1150-2.55 | 730 |
| 32DK471 | 470 | ±10 | 300 | 385 | 20 | 40 | 1250-2.55 | 780 |
| 32DK511 | 510 | ±10 | 320 | 410 | 20 | 40 | 1300-2.55 | 840 |
| 32DK561 | 560 | ±10 | 350 | 450 | 20 | 40 | 1400-2.5 | 910 |
| 32DK621 | 620 | ±10 | 385 | 505 | 20 | 40 | 1550-2.5 | 990 |
| 32DK681 | 680 | ±10 | 420 | 560 | 20 | 40 | 1700-2.5 | 1000 |
| 32DK711 | 710 | ±10 | 440 | 585 | 20 | 40 | 1750-2.45 | 1010 |
| 32DK751 | 750 | ±10 | 460 | 615 | 20 | 40 | 1800-2.45 | 1020 |
| 32DK781 | 780 | ±10 | 485 | 640 | 20 | 40 | 1950-2.45 | 1030 |
| 32DK821 | 820 | ±10 | 510 | 670 | 20 | 40 | 2000-2.40 | 1080 |
| 32DK911 | 910 | ±10 | 550 | 745 | 20 | 40 | 2100-2.3 | 1170 |
| 32DK951 | 950 | ±10 | 575 | 765 | 20 | 40 | 2200-2.3 | 1180 |
| 32DK102 | 1000 | ±10 | 625 | 825 | 20 | 40 | 2400-2.3 | 1220 |
| 32DK112 | 1100 | ±10 | 680 | 895 | 20 | 40 | 2700-2.3 | 1350 |
| 32DK122 | 1200 | ±10 | 750 | 1060 | 20 | 40 | 3000-2.3 | 1400 |

## 2. Electrical characteristics — low-residual-voltage ratings

| Model | U1mA (V) | ΔU1mA (%) | UAC (V) | UDC (V) | In (kA) | Imax (kA) | UP (V) | ETM (J) |
|---|---|---|---|---|---|---|---|---|
| 32DK431 | 430 | ±10 | 275 | 350 | 20 | 40 | 950-2.2 | 730 |
| 32DK471 | 470 | ±10 | 300 | 385 | 20 | 40 | 1030-2.2 | 780 |
| 32DK511 | 510 | ±10 | 320 | 410 | 20 | 40 | 1130-2.2 | 840 |
| 32DK561 | 560 | ±10 | 350 | 450 | 20 | 40 | 1240-2.2 | 910 |
| 32DK621 | 620 | ±10 | 385 | 505 | 20 | 40 | 1350-2.15 | 990 |
| 32DK681 | 680 | ±10 | 420 | 560 | 20 | 40 | 1480-2.15 | 1000 |
| 32DK711 | 710 | ±10 | 440 | 585 | 20 | 40 | 1540-2.15 | 1010 |
| 32DK751 | 750 | ±10 | 460 | 615 | 20 | 40 | 1630-2.15 | 1020 |
| 32DK781 | 780 | ±10 | 485 | 640 | 20 | 40 | 1670-2.15 | 1030 |
| 32DK821 | 820 | ±10 | 510 | 670 | 20 | 40 | 1850-2.1 | 1080 |
| 32DK911 | 910 | ±10 | 550 | 745 | 20 | 40 | 1940-2.1 | 1170 |
| 32DK951 | 950 | ±10 | 575 | 765 | 20 | 40 | 2040-2.1 | 1180 |
| 32DK102 | 1000 | ±10 | 625 | 825 | 20 | 40 | 2200-2.1 | 1220 |
| 32DK112 | 1100 | ±10 | 680 | 895 | 20 | 40 | 2380-2.1 | 1350 |
| 32DK122 | 1200 | ±10 | 750 | 1060 | 20 | 40 | 2620-2.1 | 1400 |

Ordering and support: samples and design-in support — North America: Thomas Dong, Flyboat Electronic LLC, Bellevue, WA (thomasd@flyboatmov.com); China: Colin Fei (colinf@flyboatmov.com). Specifications are subject to change without notice; verify critical ratings against the model row above before design freeze.